Huayi Zheng

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RESEARCH INTERESTS

- The formation of Antarctic Bottom Water
- North Atlantic heat and freshwater changes
- Contemporary sea level change

EDUCATION

Lanzhou University **Sep. 2019 - Jun. 2023**
B.S. in Atmospheric Sciences

Institute of Atmospheric Physics, Chinese Academy of Sciences/University of Chinese Academy of Sciences **Sep. 2023 - Jun. 2026**
M.S. in Atmospheric Sciences
Advisor: Dr. Lijing Cheng

British Antarctic Survey/University of Southampton **Sep. 2026 - present**
Ph.D. in Ocean and Earth Sciences
Advisor: Dr. Andrew Meijiers and Dr. Alessandro Silvano

HONORS AND AWARDS

The First Prize of 14th Student Academic Forum of Harbin Institute of Technology, Harbin Institute of Technology	2021
Outstanding Undergraduate Fellowship, Lanzhou University	2021
National Undergraduate Research Fellowship, China Ministry of Education	2022
Summer Undergraduate Research Fellowship in Marine Environmental Science, State Key Laboratory of Marine Environmental Science, Xiamen University	2022
National Scholarship, China Ministry of Education	2025
IGNITE Doctoral Landscape Award, University of Southampton	2026

PUBLICATIONS

- Pan, Y., Cheng, L., Storto, A., Mayer, M., Gulev, S., Li, Y., Lyu, K., **Zheng, H.**, & Yuan, H. (2026). Ocean Meridional Heat Transport Estimated from Energy Budget Constraint. Journal of Climate, 39, 3001–3019. https://doi.org/10.1175/JCLI-D-25-0522.1
- **Zheng, H.**, Cheng, L., Dangendorf, S., Meyssignac, B., Barnoud, A., Trenberth, K., et al. (2026). Improved closure of the global mean sea level budget from observational advances since 1960. Science Advances, 12, eaea0652. https://doi.org/10.1126/sciadv.aea0652
- Guo, J., Cheng, L., Pan, Y., **Zheng, H.**, & Ma, Z. (2026). Sensitivity of ocean heating rate estimate to time series processing methods. Accepted by Advances in Atmospheric Sciences.
- **Zheng, H.**, Cheng, L., Li, F., Pan, Y., & Zhu, C. (2024). An Observation-Based Estimate of Atlantic Meridional Freshwater Transport. Geophysical Research Letters, 51, e2024GL110021. https://doi.org/10.1029/2024GL110021
- Cheng, L., Pan, Y., Tan, Z., **Zheng, H.**, Zhu, Y., Wei, W., Du, J., Yuan, H., Li, G., Ye, H., Gouretski, V., Li, Y., Trenberth, K. E., Abraham, J., Jin, Y., Reseghetti, F., Lin, X., Zhang, B., Chen, G., & Zhu, J. (2024). IAPv4 ocean temperature and ocean heat content gridded dataset. Earth System Science Data, 16, 3517–3546. https://doi.org/10.5194/essd-16-3517-2024
- Zhang, J., **Zheng, H.**, Xu, M., Yin, Q., Zhao, S., Tian, W., & Yang, Z. (2022). Impacts of stratospheric polar vortex changes on wintertime precipitation over the northern hemisphere. Climate Dynamics, 58, 1–17. https://doi.org/10.1007/s00382-021-06088-x

SELECTED PRESENTATIONS

General Assembly 2025 of the European Geosciences Union, Vienna, April 2025 (Oral Presentation)
Sea level budget in light of recent observational advances since 1960
The Seventh Xiamen Symposium on Marine Environmental Sciences, Xiamen, January 2025 (Oral Presentation)
Sea level budget in light of recent observational advances since 1960
Joint meeting IQuOD/GTSPP/SOOP/XBT science meeting, Bologna, November 2024 (Oral Presentation)
Sea level budget in light of recent observational advances since 1960
The 7th International Ocean Salinity Science Conference, virtual, May 2024 (Oral Presentation)
An observation-based estimate of the Atlantic meridional freshwater transport from 2004 to 2012
General Assembly 2024 of the European Geosciences Union, Vienna, April 2024 (Poster)
An observation-based estimate of the Atlantic meridional freshwater transport from 2004 to 2012
Marine Multidisciplinary Cross-Integration Seminar, Xiamen, July 2022 (Oral Presentation)
Quantifying the multiple time-scales variability of salinity amplification
Asia-Oceania Geoscience Society (AOGS) 18th Annual Meeting, virtual, August 2021 (Oral Presentation)

COMMUNITY ACTIVITIES

Member, Asia Oceania Geosciences Society — Aug 2021 - Aug 2022
Member, European Geosciences Union — Apr 2024 - present

SKILLS

- Languages: Fluent - English and Native - Chinese
- Computer: Python, NCL, Fortran, LATEX